

CofactGVR: Counterfactual Intervention for Grounded Visual Reasoning

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Abstract

Despite rapid progress in Grounded Visual Reasoning (GVR) with MLLMs and RL-style fine-tuning, existing approaches often lack effective learning signals for intermediate grounding decisions and are prone to shortcut solutions. In this work, we explicitly decompose GVR into Evidence Generation followed by Counterfactual Answer Reasoning, and formalize this structure as a Causal Grounding Graph (CGG) in which the generated evidence acts as a causal mediator. Building on this formulation, we propose CofactGVR, which estimates the mediator’s utility via a matched counterfactual intervention that perturbs the predicted region while keeping the original image–question context fixed. The factual–counterfactual reward gap yields a principled intermediate bonus, selectively assigned to high-quality factual rollouts to promote evidence-faithful reasoning. To further stabilize and efficiently exploit this causal training signal, we incorporate a Quantile-filtered Prioritized Advantage Sampling (Q-PAS) strategy that preferentially updates on trajectories with high-magnitude advantages while filtering low-signal samples. Extensive experiments across GVR benchmarks show consistent improvements, indicating that CofactGVR strengthens reliance on informative visual evidence under controlled interventions. Our source code will be released at <https://github.com/zhangy0822/CofactGVR>.

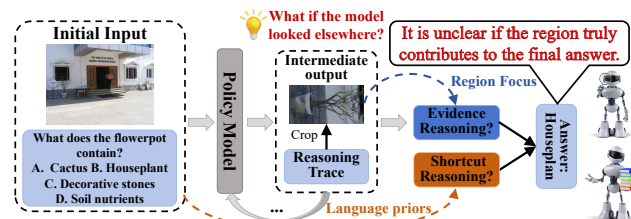


Figure 1. **Motivation: Causal ambiguity in GVR.** It is often indistinguishable whether an answer stems from faithful evidence reasoning or merely exploits shortcuts (e.g., language priors). We address this by asking: “What if the model looked elsewhere?” By employing **counterfactual intervention**, we quantify the evidence utility—the causal performance gap that reveals whether the model truly relies on the visual evidence.

1. Introduction

Grounded Visual Reasoning (GVR) has recently emerged as a pivotal testbed for multimodal intelligence, requiring a model to not only answer a question about an image but also ground its reasoning in explicit visual evidence, typically in the form of bounding boxes (Cao et al., 2025; Zheng et al., 2026; Zhang et al., 2025a; Wang et al., 2026). Propelled by Multimodal Large Language Models (MLLMs) (Liu et al., 2023; Zhang et al., 2024c; Bai et al., 2025) and Reinforcement Learning (RL) fine-tuning (Shao et al., 2024b; Guo et al., 2025), modern systems have achieved remarkable progress on grounding-centric benchmarks. However, notwithstanding these performance gains, current approaches remain constrained by two limitations. First, they provide **insufficient supervision for intermediate grounding decisions**—models are typically optimized solely for final accuracy and format compliance, while the causal utility of intermediate evidence (e.g., the predicted bounding box) is only implicitly reflected. Second, they remain **susceptible to shortcut solutions**, frequently generating plausible answers that exploit language priors or superficial cues, rather than faithfully leveraging the inferred visual evidence.

As exemplified in Figure 1, these limitations are clearly manifested, yet the training feedback fails to discern whether the final answer stems from faithful evidence reasoning or shortcut reasoning. To address this causal ambiguity, we propose a counterfactual intuition: **if a model truly relies on the visual evidence, its ability to predict the correct answer should be significantly impaired if we force the**

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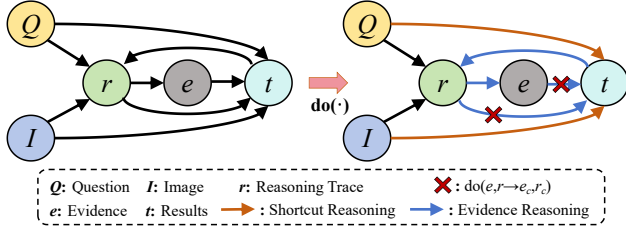


Figure 2. **Illustration of the Causal Grounding Graph (CGG).** (Left) The observational graph. (Right) The interventional graph under the $\text{do}(\cdot)$ operator. This intervention severs natural dependencies to decouple faithful Evidence Reasoning from Shortcut Reasoning, enabling the quantification of causal evidence utility.

model to “look elsewhere” (i.e., at an irrelevant region).

By quantifying the performance gap between looking at the predicted evidence and looking elsewhere, we can measure the true utility of the grounded evidence.

To address these challenges, this paper argues that GVR necessitates treating visual evidence as a crucial decision variable with explicit bonus assignment—a critical perspective largely overlooked by prior works (Fu et al., 2025b; Cao et al., 2025; Sarch et al., 2025). We therefore decompose GVR into a two-stage process: (i) **Evidence Generation**: inferring visual evidence (e.g., bounding boxes) conditioned on the image and question; and (ii) **Counterfactual Answer Reasoning**: performing comparative reasoning that contrasts the factual evidence-induced view against a matched counterfactual view, derived via intervention on the predicted box. Formally, we formalize this decomposition as a **Causal Grounding Graph (CGG)**. As depicted in Figure 2, the image I and question Q act as the initial inputs; r represents the reasoning trace output by the Evidence Generation stage, which in turn yields the visual evidence e ; and the final answer t is generated based on $\{Q, I, r, \text{ and } e\}$.

Building on this formulation, we propose CofactGVR, a counterfactual bonus shaping framework that explicitly attributes reward to grounding visual evidence via counterfactual intervention. Concretely, for each instance, we first sample multiple rollouts from a policy model in the Evidence Generation stage, obtaining a set of reasoning traces and corresponding visual evidence that instantiate evidence-specific views. To quantify the contribution of factors other than visual evidence to the final results t , we employ the *do*-operator ($\text{do}(\cdot)$) (Glymour et al., 2016; Pearl, 2018; Wang et al., 2020; Zhang et al., 2023) to eliminate the influence of visual evidence, as illustrated in Figure 2. Specifically, in the Counterfactual Answer Reasoning stage, we construct a matched counterfactual branch by intervening on the visual evidence (depicted as $\times$ in Figure 2), which replaces the predicted box with a randomized perturbation and keeps the image–question context fixed. This design isolates the contribution of evidence: if the factual branch substantially outperforms its counterfactual counterpart, the predicted

evidence is deemed causally informative; conversely, if the reward remains similar, the model generates the final answer through shortcut reasoning that is weakly dependent on the visual evidence. Then, CofactGVR translates this causal contrast into a selective intermediate learning signal. Specifically, we estimate an evidence utility score from the factual–counterfactual reward gap and convert it into a bonus applied exclusively to high-quality factual rollouts. This framework effectively addresses the above limitations by explicitly distinguishing the sources of the reward, thereby determining whether the generated answer stems from faithful visual evidence or shortcut correlations.

A practical challenge arises from the inherently stochastic nature of RL optimization in MLLMs, where the causal gap signal often exhibits sparsity and high variance across sampled rollouts. To stabilize optimization and enhance sample efficiency, we incorporate a Quantile-filtered Prioritized Advantage Sampling (Q-PAS) strategy, which preferentially targets trajectories with substantial advantage magnitudes while pruning low-signal samples. Under the CGG framework, the counterfactual bonus induces a causal advantage signal whose informative mass is concentrated within a small subset of trajectories. Q-PAS precisely amplifies these high-signal instances, making the causal supervision more tractable and stable throughout training.

The main contributions of this paper are summarized as follows:

- We explicitly decompose GVR into a two-stage process—Evidence Generation followed by Counterfactual Answer Reasoning, and formalize GVR as a Causal Grounding Graph (CGG) where the generated visual evidence acts as a causal mediator in grounding dialogue.
- We propose CofactGVR, which systematically integrates counterfactual intervention into MLLM-based reasoning, and transforms reward gap into a selective bonus assigned for the intermediate reasoning.
- We integrate Quantile-filtered Prioritized Advantage Sampling (Q-PAS) to stabilize and efficiently exploit the causal training signal by emphasizing high-advantage trajectories and filtering low-signal samples.
- Extensive experiments across GVR benchmarks show consistent gains and more faithful visually grounded reasoning under controlled interventions.

2. Related Work

2.1. Grounded Visual Reasoning

Grounded visual reasoning (GVR) aims to enable model not only answer correctly but also justify their decisions by

localizing task-relevant visual evidence. Prior work couples reasoning with region-level perception, ranging from box-based region feature modeling (e.g., GPT4RoI (Zhang et al., 2024b)) to grounding via large-scale aligned image-text corpora (e.g., Kosmos-2 (Peng et al., 2024)). Other methods further introduce visual search or tool-augmented pipelines to acquire localized evidence for downstream reasoning (Liu et al., 2024d; Peng et al., 2024; Wu & Xie, 2024; Li et al., 2025b; Khayatkhoei et al., 2025), albeit with additional system complexity and reliance on external tools. Recent approaches increasingly interleave evidence acquisition with chain-of-thought style reasoning, such as VisCoT (Shao et al., 2024a), VoCoT (Li et al., 2025b), CogCoM (Qi et al., 2025), ViGoRL (Sarch et al., 2025), and FAST (Sun et al., 2025), which improve interpretability and accuracy on evidence-heavy benchmarks but typically require annotated evidence regions. In contrast, Ground-R1 (Cao et al., 2025) shows that reinforcement learning can elicit grounding behaviors without bounding-box supervision. Building on this direction, our work focuses on *making generated evidence causally useful* for answering, and introduces counterfactual supervision to mitigate shortcut reasoning and strengthen evidence dependence.

2.2. Counterfactual Learning for MLLMs

Counterfactual learning aims at improving the faithfulness of MLLMs by introducing the causal discovery and explicitly contrasting the predictions with controlled counterfactual variants (Zhang et al., 2024a; Li et al., 2025a). This line of work is closely related to broader causal learning themes, including causal discovery (Li et al., 2025a) and causal representation learning (Ahuja et al., 2023; Li et al., 2024). Recent studies further explore how MLLMs and LLMs assist or even directly infer causal relations from unstructured descriptions (Liu et al., 2024a; Li et al., 2025a; Vashishtha et al., 2025). A representative family of approaches performs training-free contrastive decoding by comparing logits from the original input against an auxiliary distribution constructed from counterfactual views (Li et al., 2023b; Chuang et al., 2024). Training-based methods construct specialized counterfactual datasets via prompting, filtering, or annotation with expensive data curation cost (Gunjal et al., 2024; Liu et al., 2024b). Overall, existing methods largely operate at the input level or via data synthesis to mitigate spurious correlations. In contrast, GVR produces the final answer through an evidence-centric process, where explicit visual evidence and intermediate reasoning traces serve as key intermediates. Building on this perspective, our work brings counterfactual learning into the GVR pipeline: instead of perturbing the raw input or relying on newly synthesized counterfactual data, we construct counterfactual rollouts that directly intervene on the predicted evidence. We further convert the resulting factual-counterfactual up-

lift into a training signal via *counterfactual bonus shaping*, encouraging the model to produce evidence that is not only plausible but also *causally useful* for answering.

3. Methodology

3.1. Overview

Our proposed CofactGVR method consists of Evidence Generation and Counterfactual Answer Reasoning stages, as illustrated in Figure 3. Unlike Ground-R1 (Cao et al., 2025), although it includes the grounding phase and the answering phase, the single prompt employed leads to the final policy converging to a single-stage generation approach rather than leveraging the generated visual evidence. Please refer to the Figure 6 for the prompts of the two stages. Given a training instance $x = (I, Q, y)$ consisting of an image I , a question Q , and the ground-truth answer y , CofactGVR samples two matched groups of rollouts under the same policy backbone: i) employs the predicted grounding box to crop the image and perform grounded reasoning, and ii) replaces the predicted box with a random box (paired with a corrupted/noise cropped image) as a controlled baseline.

3.2. Causal Grounding Graph

To explicitly quantify and incentivize the causal contribution of grounding, we introduce a Causal Grounding Graph (CGG) over the variables $\{I, Q, r, e, t\}$, where r denotes the textual reasoning trace of Evidence Generation stage, e denotes the cropped visual evidence region, and t is the final answer. As shown in Figure 3, we argue GVR involves two information pathways: an *shortcut reasoning* path where the model predicts the answer directly from (I, Q) or r , and a *evidence reasoning* path where the model generates an answer based on the grounding evidence e obtained from the Evidence Generation stage.

As shown in the upper left corner of Figure 3, CGG encodes the structural dependencies: $(I, Q) \rightarrow r$, $r \rightarrow e$, $(I, Q) \rightarrow t$, $(r, e) \rightarrow t$. The final answer t is influenced both by the shortcut reasoning path $(I, Q) \rightarrow t$ and the evidence reasoning path $(r, e) \rightarrow t$. Our goal is to encourage policies whose performance gains are attributable to the grounded path rather than shortcut cues. To achieve this, we apply the *do*-operator (Glymour et al., 2016; Pearl, 2018) to perform a structural intervention on the visual evidence and the reasoning trace, denoted as $do(e, r \rightarrow e_c, r_c)$, which are represented as red $\times$ in Figure 3. Please refer to Section 3.4 for specific procedure.

3.3. Evidence Generation

The Evidence Generation stage samples a group of candidate evidence boxes $\mathcal{B} = \{b_i\}_{i=1}^E$ and corresponding reasoning

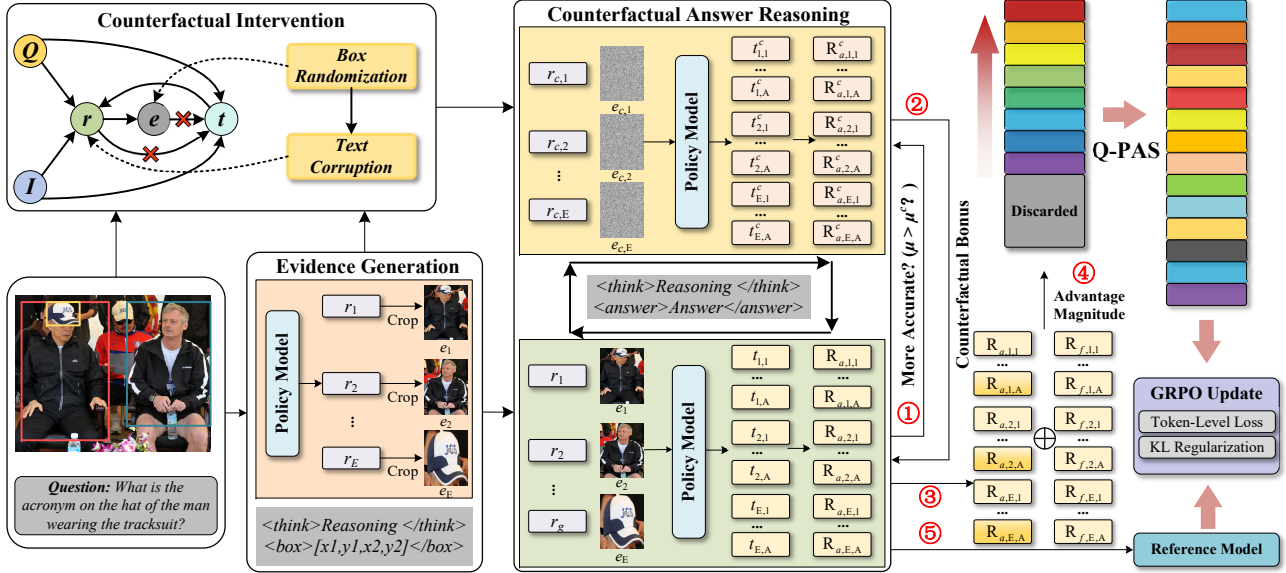


Figure 3. Overview of CofactGVR. In the Evidence Generation stage, the policy generates the reasoning trace r_i and grounding visual evidence e_i . In the Counterfactual Answer Reasoning stage, the policy generates the counterfactual answer $t_{i,j}^c$ based on counterfactual evidence $e_{c,i}$ and the factual answer $t_{i,j}$ based on factual evidence e_i . ①: Compute the corresponding average accuracy rewards μ and μ^c , and estimate the evidence utility via a factual-counterfactual reward gap $\mu - \mu^c$. ②: Convert the gap into a selective bonus that is assigned to high-quality factual rollouts. ③: Combine the accuracy reward and the format reward. ④: Apply Quantile-filtered Prioritized Advantage Sampling (Q-PAS) to prioritize updates on trajectories with large-magnitude advantages while filtering low-signal samples. ⑤: Input the context to the Reference Model for obtaining the KL divergence penalty. Note that the second stage supports multiple rounds of reasoning, and the above illustration is an example of obtaining answers directly.

trace $\mathcal{R} = \{r_i\}_{i=1}^E$ from the current rollout policy $\pi_{\theta_{\text{old}}}$, where $b_i \in \mathbb{R}^4$ denotes the axis-aligned bounding box represented by its top-left and bottom-right corner coordinates. Then, we extract the corresponding evidence regions via a crop operator $\text{Crop}(\cdot)$: $e_i = \text{Crop}(I, b_i)$, $i \in [1, E]$.

3.4. Counterfactual Answer Reasoning

Given the predicted box b_i predicted in the first stage, we perform the standard grounded answer reasoning by cropping the corresponding region and generating answer rollouts under the same policy backbone. The answer rollouts $f \sim \pi_{\theta_{\text{old}}}(\cdot | I, Q, e_i)$ of the factual path are omitted for brevity. To explicitly measure the causal utility of the predicted evidence, CofactGVR introduces a matched counterfactual reasoning branch as a controlled baseline. Specifically, we replace the predicted box with a random box $\tilde{b} \sim p_{\text{rand}}(\cdot | I)$ and apply an intervention operator $\mathcal{I}(\cdot)$ (e.g., noise corruption) on the cropped region, yielding the counterfactual evidence $e_{c,i} = \mathcal{I}(\text{Crop}(I, \tilde{b}))$. This intervention effectively forces the model to “look elsewhere”, serving as a counterfactual baseline to verify whether the original gaze was necessary. Meanwhile, since the reasoning trace r_i explicitly contains the predicted box token, we enforce consistency by replacing the original box text in r_i with $\tilde{b}$, yielding the counterfactual trace: $r_{c,i}$.

This coupled intervention $(r_i, e_i) \rightarrow (r_{c,i}, e_{c,i})$ blocks

the grounded causal pathway while avoiding mismatched cues between textual coordinates and visual crops, making the factual-counterfactual reward gap a faithful proxy of grounding utility. The policy then generates counterfactual answer rollouts as follows:

$$c_i = \{c_{i,j}\}_{j=1}^A \sim \pi_{\theta_{\text{old}}}(\cdot | I, Q, e_{c,i}), i \in [1, E], \quad (1)$$

where $c_{i,j}$ denote the j -th rollout answers based on the i -th counterfactual evidence region $e_{c,i}$. A represents the number of answer rollouts generated in the counterfactual answer reasoning stage. Overall, CofactGVR generates a total of $G = E \times A$ answers. This design ensures that the only difference between the two branches lies in the evidence intervention, making the reward gap a faithful proxy of evidence usefulness for grounded reasoning.

3.5. Counterfactual Bonus Shaping

CofactGVR assigns an additional counterfactual bonus to factual rollouts, ensuring that the policy is incentivized solely when the predicted evidence confers a causal uplift over a corrupted counterfactual baseline. Let $\{R_{a,i,j}\}_{i,j=1}^{E,A}$ and $\{R_{a,i,j}^c\}_{i,j=1}^{E,A}$ denote the accuracy rewards for the sets of $G = E \times A$ factual and counterfactual answer rollouts, respectively. We commence by calculating the group-level

average accuracy reward:

$$\mu = \frac{1}{G} \sum_{i=1}^E \sum_{j=1}^A R_{a,i,j}, \quad \mu^c = \frac{1}{G} \sum_{i=1}^E \sum_{j=1}^A R_{a,i,j}^c, \quad (2)$$

and obtain the uplift by comparing the mean accuracy reward: $d = [\mu - \mu^c]_+$, where $[\cdot]_+ = \max(\cdot, 0)$ enforces that the bonus is activated only when the factual branch outperforms the counterfactual baseline. To avoid vanishing shaping signals while preventing overly aggressive reward inflation, we apply a trigger-and-clip strategy to construct a bounded counterfactual bonus:

$$b = \mathbb{I}(d > \gamma) \cdot \text{clip}(d, b_{\min}, b_{\max}), \quad (3)$$

where γ is a small threshold, and $b_{\min}/b_{\max}$ denote the bonus floor and cap, respectively.

Rollout-level Gating. Instead of uniformly assigning b to all factual rollouts, CofactGVR introduces a rollout-level gating mask to concentrate the bonus on reliable rollouts. We first attempt a correctness-based mask using a confidence threshold τ for questions with discrete answer types, such as multiple-choice questions:

$$m_{i,j}^{\text{corr}} = \mathbb{I}(R_{a,i,j} \geq \tau), \quad (4)$$

where the accuracy rewards usually consist of only 0 or 1. However, in a majority of scenarios, many rewards are not discrete but rather continuous, such as leveraging ROUGE-1, ROUGE-2, and ROUGE-L as reward. Thus, we further introduce $\mathbb{I}_{\text{has}}$ to indicate whether at least one factual rollout is confidently correct:

$$\mathbb{I}_{\text{has}} = \mathbb{I}\left(\sum_{i=1}^E \sum_{j=1}^A m_{i,j}^{\text{corr}} > 0\right), \quad (5)$$

where $\mathbb{I}_{\text{has}}$ indicates whether at least one factual rollout is confidently correct. If no such rollout exists, we fall back to a top- k mask that selects the k rollouts with the highest factual accuracy rewards:

$$m_{i,j}^{\text{topk}} = \mathbb{I}[(i, j) \in \text{Topk}(R_{a,u,v} \mid u \in [1, E], v \in [1, A])], \quad (6)$$

where k is a small constant (e.g., $k = 3$). The final gating mask is defined as

$$m_{i,j} = \mathbb{I}_{\text{has}} \cdot m_{i,j}^{\text{corr}} + (1 - \mathbb{I}_{\text{has}}) \cdot m_{i,j}^{\text{topk}}. \quad (7)$$

Final Accuracy Reward. Finally, we shape the factual accuracy reward by adding the bounded counterfactual bonus to the gated rollouts:

$$\tilde{R}_{a,i,j} = R_{a,i,j} + b \cdot m_{i,j}, \quad i \in [1, E], \quad j \in [1, A]. \quad (8)$$

The shaped rewards $\tilde{R}_{a,i,j}$ are then adopted in subsequent advantage computation and policy optimization. This design encourages evidence selection that yields non-trivial causal gains over counterfactual interventions, while avoiding indiscriminate reward boosting on low-quality rollouts.

3.6. GRPO Optimization

With the counterfactually shaped rewards $\{\tilde{R}_{a,i,j}\}_{i,j=1}^{E,A}$ defined in Equation (8), we optimize the policy using GRPO (Shao et al., 2024b) with a Q-PAS strategy to improve sample efficiency and training stability. Although the counterfactual bonus shaping operates on the accuracy term, the final reward employed for policy optimization additionally includes a format-related constraint. Concretely, for each rollout we define the composite reward:

$$R_{i,j} = \tilde{R}_{a,i,j} + \lambda_f R_{f,i,j}, \quad i \in [1, E], \quad j \in [1, A], \quad (9)$$

where $\tilde{R}_{a,i,j}$ is the accuracy reward after counterfactual shaping (Eq. 8), $R_{f,i,j}$ is the format reward computed on the factual completion, and λ_f balances the two terms. In our implementation, the counterfactual branch is used only to estimate the causal uplift signal, while optimization is performed on factual rollouts with $R_{i,j}$.

Quantile-filtered Prioritized Advantage Sampling. Acknowledging the non-uniform contribution of rollouts to policy improvement, we preferentially target trajectories with significant advantage magnitudes, thereby filtering out low-signal samples. Specifically, we compute the group-wise advantage $A_{i,j}$ (Please refer to Section A.1) and compute a discarded threshold:

$$\omega = \text{Quantile}(\{|A_{i,j}|\}_{i,j=1}^{E,A}, q), \quad (10)$$

where $q \in (0, 1)$ controls the filtered ratio. The selected subset is $\mathcal{S} = \{A_{i,j} \mid |A_{i,j}| \geq \omega\}$. This mechanism concentrates optimization on rollouts that exhibit strong learning signals (either positive or negative), while improving robustness against noisy samples. Differently, ReVisual-R1 (Chen et al., 2025) keeps only a middle-dominance band, which is often uninformative under low within-group dominance variation and thus fails to identify effective samples. Then, we define a prioritized sampling distribution $p(g) = s_g^\alpha / \sum_{t \in \mathcal{E}} s_t^\alpha$, $g \in \mathcal{S}$, where $\alpha \geq 0$ denotes the prioritization exponent that controls how strongly sampling concentrates on large-magnitude advantages. We then draw $\rho \times G$ indices with replacement, where ρ is the sampling ratio:

$$\mathcal{M} = \{g_m\}_{m=1}^M \sim p(g), \quad g_m \in \mathcal{E}. \quad (11)$$

Finally, we compute the GRPO objective on the resampled subset. This Q-PAS procedure concentrates optimization on rollouts with larger $|A|$, thereby improving gradient signal-to-noise ratio and stabilizing training under counterfactual interventions. Please refer to Algorithm 1 for a more detailed explanation.

Table 1. Comparison with SoTA methods on the **test split of VisCoT benchmark**, including Doc/Text and Chart understanding, General VQA, Relation Reasoning, and Fine-Grained VQA.

Method	Doc/Text					Chart	General VQA		Relation Reasoning			FGVQA
	DocVQA	TextCaps	TextVQA	DUDE	SROIE	InfoVQA	Flickr30k	Visual7W	GQA	Open images	VSR	CUB
LLaVA-1.5-7B (2024c)	24.4	59.7	58.8	29.0	13.6	40.0	58.1	57.5	53.4	41.2	57.2	53.0
LLaVA-1.5-13B (2024c)	26.8	61.5	61.7	28.7	16.4	42.6	62.0	58.0	57.1	41.3	59.0	57.3
SPHINX-13B (2023)	19.8	55.1	53.2	0.0	7.1	35.2	60.7	55.8	58.4	46.7	61.3	50.5
VisCoT-7B (2024a)	47.6	67.5	77.5	38.6	47.0	32.4	66.8	55.8	63.1	82.2	61.4	55.9
CogCoM (2025)	–	–	71.1	–	–	–	–	–	71.7	–	–	–
Chain of Spot (2024f)	–	–	60.9	–	–	–	–	–	63.7	–	–	–
FAST (2025)	–	–	60.7	–	–	–	–	–	63.8	–	–	–
Vision-R1-7B (2026)	86.0	71.9	88.3	67.0	89.9	61.3	64.4	47.2	79.0	74.0	67.0	81.3
LMM-R1 (2025)	84.3	78.0	91.2	66.7	78.5	55.6	61.7	52.6	80.0	76.7	40.2	64.8
Ground-R1 (2025)	88.4	79.7	95.3	68.3	89.0	69.0	60.1	64.3	90.0	83.3	76.3	84.7
CofactGVR	92.6	80.3	96.3	72.1	89.9	70.3	71.2	64.8	91.3	87.7	77.0	85.0

4. Experiments

4.1. Experimental Setup

Our CofactGVR model is trained on the VisCoT dataset (Shao et al., 2024a), which contains large-scale visual question-answer pairs accompanied by bounding-box annotations highlighting key evidence regions. Following the method of Ground-R1 (Cao et al., 2025), we employ only 8K samples, while all bounding-box annotations are discarded during training. We evaluate our method on a comprehensive suite of benchmarks. For VQA, we employ the test split of VisCoT (Shao et al., 2024a) and the established LVLM benchmarks, including MME (Fu et al., 2025a), SEED-Bench (Li et al., 2023a), MME-RealWorld-Lite (Zhang et al., 2025b), RealworldQA (xAI, 2024), and POPE (Li et al., 2023c). For visual grounding, we report results on RefCOCO+/g (Kazemzadeh et al., 2014; Mao et al., 2016). Note that the counterfactual bonus shaping is exclusively employed during training and is discarded at test time. Thus, CofactGVR incurs no additional inference latency compared with standard two-stage grounding pipelines. Implementation details are provided in Section A.2.

4.2. Main Results

We present the main quantitative results on three aspects: (i) VisCoT test split, which contains diverse VQA settings, (ii) general LVLM benchmarks for broader multimodal robustness, and (iii) visual grounding benchmarks for referring expression localization. As shown in Table 1–Table 3, we compare CofactGVR with representative LVLM baselines.

Evaluation on VisCoT. We report the results of test split of the VisCoT in Table 1, which covers five reasoning tiers including doc/text understanding, chart reasoning, general VQA, relation reasoning, and fine-grained VQA. Overall, our method achieves the best performance on all 12 tasks, demonstrating consistent gains over strong baselines and recent R1-style models. Compared with Ground-R1, our model yields notable improvements on document-centric tasks, e.g., DocVQA (+4.2), DUDE (+3.8), and TextVQA

Table 2. Comparison with SoTA methods on **LVLM benchmarks**.

Method	MME	SEED	MME-RWL	RWQA	POPE
GPT-4V (2023)	1926.6	53.8	–	61.4	–
GPT-4o (2024)	–	72.0	–	75.4	86.9
InternVL2-8B (2024b)	2210.3	–	–	64.4	86.9
InternVL2.5-8B (2024a)	2344.1	–	–	70.1	90.6
Qwen2-VL-7B (2024b)	2326.8	75.1	–	70.1	88.1
Chain of Spot (2024f)	1501.1	59.7	–	–	86.4
V* (2024)	1128.9	41.7	–	–	82.4
FaST (2025)	1517.4	60.1	–	–	86.4
Vision-R1-7B (2026)	2306.2	73.0	41.0	50.2	88.7
LMM-R1 (2025)	2210.6	74.4	35.4	52.2	86.5
R1-Onevision (2025)	2192.2	66.5	35.1	46.5	84.9
Ground-R1 (2025)	2355.6	74.8	38.3	65.4	88.4
VGR (2026)	–	–	–	59.8	88.2
CofactGVR	2428.5	76.1	42.4	66.9	90.0

(+1.0). We also observe substantial gains on general VQA and relation reasoning, especially on Flickr30k (+11.1), Open Images (+4.4), and GQA (+1.3), indicating stronger generalization beyond evidence-heavy settings.

Evaluation on LVLM benchmarks. As shown in Table 2, CofactGVR consistently improves over Ground-R1 across all reported benchmarks, including MME (+72.9), SEED (+1.3), and MME-RWL (+4.1). These gains, alongside improvements on RWQA (+1.5) and POPE (+1.6), indicate that our counterfactual shaping not only benefits grounded reasoning accuracy but also improves real-world robustness and reduces hallucination tendencies.

Visual Grounding Results. We further evaluate referring expression grounding on RefCOCO, RefCOCO+ and RefCOCOg in Table 3. Our model achieves the best results on RefCOCO/RefCOCO+, with particularly strong improvements on RefCOCO+ (e.g., +3.6 on val, +4.1 on test-A, and +6.5 on test-B) compared with Ground-R1. On RefCOCOg, we observe a small drop (−1.1 on val and −0.6 on test), likely stems from the prioritizing of visual evidence slightly impacts the processing of linguistically complex, long-form expressions. These results indicate that CofactGVR strengthens grounding capability on challenging splits (especially RefCOCO+) while maintaining strong performance on long-expression grounding benchmarks.

Table 3. Comparison with SoTA methods on **RefCOCO**, **RefCOCO+**, and **RefCOCOg**.

Method	RefCOCO			RefCOCO+			RefCOCOg	
	val	test-A	test-B	val	test-A	test-B	val	test
Gemini 1.5 Pro (2024)	73.2	72.9	74.6	62.5	63.9	65.0	75.2	76.2
Grounding DINO (2024e)	90.6	93.2	88.2	88.2	89.0	75.9	86.1	87.0
Shikra-7B (2023)	87.0	90.6	80.2	81.6	87.4	72.1	82.3	82.2
Shikra-13B (2023)	87.8	91.1	81.8	82.9	87.8	74.4	82.6	83.2
Qwen-VL (2023)	89.4	92.3	85.3	83.1	88.3	77.2	85.6	85.5
CogVLM (2024c)	92.5	94.0	88.7	87.5	91.8	81.4	89.5	90.1
CogCoM (2025)	92.3	94.6	89.2	88.2	92.8	82.1	89.3	90.5
Qwen2.5-VL-7B (2025)	90.0	92.5	85.4	84.2	89.1	76.9	87.2	87.2
InternVL2.5-8B (2024a)	90.3	94.5	85.9	85.2	91.5	78.8	86.7	87.6
Vision-R1-7B (2026)	58.3	64.2	48.5	50.9	61.1	38.5	51.9	52.6
LMM-R1 (2025)	87.3	91.4	83.6	79.2	87.1	72.9	84.1	84.1
R1-Onevision (2025)	45.0	51.6	35.3	47.2	57.0	37.1	53.0	48.2
Ground-R1 (2025)	92.9	93.9	88.0	86.5	90.8	78.8	90.1	90.2
CofactGVR	94.0	96.6	90.9	90.1	94.9	85.3	89.0	89.6

4.3. Ablation Studies

Effect of Counterfactual Intervention and Q-PAS. As shown in Table 4, Full CofactGVR performs best, and removing any component consistently degrades results. Disabling counterfactual intervention causes the largest drop (MME/RWQA/POPE/SEED: 2428.5/66.9/90.0/76.1 $\rightarrow$ 2349.1/63.8/87.9/75.1), confirming that the counterfactual branch provides the crucial causal contrast to discourage shortcut reasoning. Removing Q-PAS also reduces performance (e.g., MME 2428.5 $\rightarrow$ 2406.2; RWQA 66.9 $\rightarrow$ 65.8), indicating that filtering low-signal rollouts and prioritizing informative ones improves optimization stability.

Designs of Counterfactual Bonus Shaping. We ablate key design choices of counterfactual bonus shaping. Removing the **bonus boundaries** causes a substantial drop (e.g., MME 2428.5 $\rightarrow$ 2341.6; SEED 76.1 $\rightarrow$ 73.7), indicating that unbounded uplift-based shaping exacerbates reward noise and destabilizes optimization. Similarly, assigning the bonus to all factual rollouts without gating markedly degrades performance, especially on RWQA (66.9 $\rightarrow$ 61.7), suggesting that indiscriminate shaping contaminates learning with low-quality trajectories. We further compare gating strategies. The *correctness-based mask only* (Equation (4)) performs poorly, consistent with the prevalence of continuous or soft correctness signals that render threshold triggers unreliable. The *top-k mask only* (Equation (6)) also underperforms, implying that ranking-only selection can be sensitive when reward distributions are flat. Overall, the bounded-and-gated design (Equation (7)) yields the most reliable gains by stabilizing bonus magnitudes via clipping and restricting bonus propagation to high-quality rollouts. Note that the rollout gating, correctness-based masking, and boundary bonuses are specific implementation details within the counterfactual intervention.

Ablation on Q-PAS. As reported in Table 4, Q-PAS consistently yields the best overall performance across four LVLM benchmarks. Compared with using *all samples*, Q-

Table 4. Ablation studies on **model architecture**, **counterfactual bonus shaping designs**, **Q-PAS variants**, and the **sensitivity analysis of prioritization exponent**.

Variants	MME	RWQA	POPE	SEED
Full CofactGVR	2428.5	66.9	90.0	76.1
Model Architecture:				
w/o counterfactual intervention	2349.1	63.8	87.9	75.1
w/o Q-PAS	2406.2	65.8	88.8	75.7
Counterfactual Bonus:				
w/o boundaries bonus	2341.6	64.1	88.7	73.7
w/o rollout gating (bonus to all)	2401.4	61.7	88.6	73.8
Correctness-based mask only	2328.9	60.8	88.3	73.9
Top- k mask only ($k = 3$)	2302.5	58.7	87.9	75.3
Q-PAS Variants:				
w/o filter (resample only)	2403.5	63.8	89.4	74.3
w/o resample (all samples)	2406.2	65.8	88.3	75.5
Prioritization Exponent α:				
$\alpha = 0$	2372.7	63.3	89.2	75.8
$\alpha = 0.5$	2428.5	66.9	90.0	76.1
$\alpha = 1.0$	2389.3	64.6	89.3	75.0
$\alpha = 2.0$	2396.7	65.1	89.3	75.2

Table 5. Ablation studies on **different counterfactual baselines**.

Variants	MME	SEED	MME-RWL	RWQA	POPE
No-evidence	2306.0	74.9	34.2	63.3	88.4
Plausible candidate region	2386.4	74.1	23.4	65.5	89.2
Full CofactGVR	2428.5	76.1	42.4	66.9	90.0

PAS improves MME from 2406.2 to 2428.5 and brings consistent gains on RWQA/POPE/SEED, indicating that selectively emphasizing informative rollouts is beneficial for stable policy improvement. Moreover, *resample only* underperforms both *all samples* and Q-PAS, suggesting that advantage-based resampling alone could be noisy when low-signal rollouts dominate the candidate pool. By filtering out low-advantage samples before prioritization, Q-PAS reduces gradient noise and avoids spending optimization budget on uninformative rollouts, leading to stronger and more consistent improvements.

We further ablate the prioritization exponent α . When $\alpha = 0$, sampling becomes effectively uniform over the filtered set $\mathcal{S}$ and performance drops notably on MME and RWQA, implying that prioritization is necessary to focus learning on high-signal rollouts. However, overly aggressive prioritization (e.g., $\alpha \geq 1.0$) also degrades performance, likely because it over-concentrates probability mass on a small subset of rollouts and harms sample diversity. Overall, $\alpha = 0.5$ provides the best trade-off between emphasizing informative rollouts and preserving diversity.

Comparison of Different Counterfactual Baselines. To address whether our noise-evidence intervention overestimates utility, we compare it against the “No-evidence” and “Plausible candidate region” baselines, as shown in Table 5.

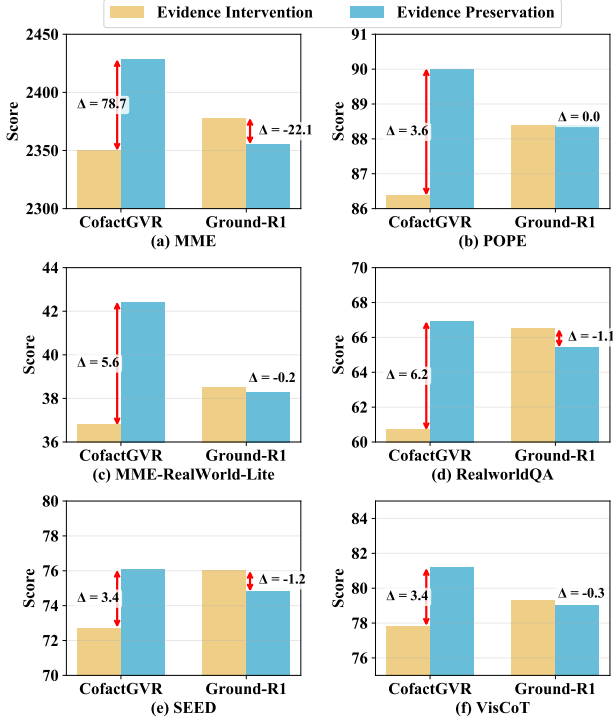


Figure 4. **Evidence Preservation vs. Evidence Intervention:** Verifying the causal utility of generated evidence.

From the results, we conclude the following conclusions. First. Plausible region introduces confounders: An irrelevant but plausible crop (e.g., a “dog” instead of a “cat”) acts as an adversarial distractor. The reward gap would measure vulnerability to misleading semantics rather than the genuine marginal contribution of the true evidence. Second. No-evidence breaks structural consistency: Removing the crop forces the model to rely entirely on the global image I and language prior Q . This degrades the comparison to multi-step vs. single-step grounding, failing to isolate the specific utility of local evidence. Third. Our Noise-evidence satisfies the requirements of a clean causal intervention: It maintains the prompt’s structural format with empty semantic and does not actively mislead the model, which isolates the “benefit of looking at the right place,” yielding a mathematically clean reward gap.

4.4. Qualitative Results and Analyses

Causal Utility of Generated Evidence. To verify whether a model *causally* relies on its generated evidence (rather than answering from shortcut reasoning cues), we conduct an evidence effectiveness analysis on several benchmarks. As shown in Figure 4, we evaluate each method under two controlled settings during inference: (i) Evidence Intervention: the predicted evidence is replaced by a randomized evidence input (e.g., randomly generated box and its corresponding reasoning trace), and (ii) Evidence Preservation: the pre-

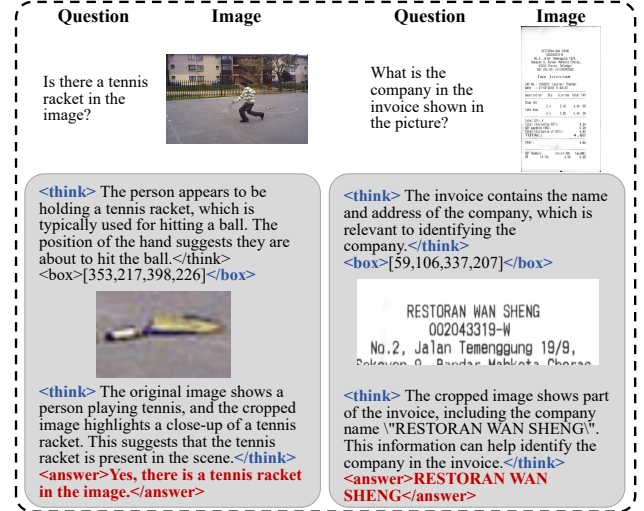


Figure 5. Reasoning trajectories of our CofactGVR on Flickr30K (left) and SROIE (right) benchmarks.

dicted evidence is kept unchanged. The performance gap Δ between the two settings serves as a proxy of the causal utility of evidence: a larger positive Δ indicates that preserving the evidence yields substantial benefits over a randomized baseline, implying stronger evidence dependence.

All the quantitative results compared with Ground-R1 (Cao et al., 2025) are shown in Figure 4. CofactGVR exhibits a consistently large and positive uplift when evidence is preserved, with $\Delta = 78.7$ on MME, $\Delta = 3.3$ on POPE, $\Delta = 5.6$ on MME-RealWorld-Lite, $\Delta = 6.2$ on RealworldQA, $\Delta = 3.4$ on SEED, and $\Delta = 3.4$ on test split of the VisCoT benchmark. The sharp performance decline under randomization confirms CofactGVR’s causal reliance on evidence. In contrast, Ground-R1 shows limited or even negative gains from evidence preservation (e.g., $\Delta = -22.1$ on MME, $\Delta = 0.0$ on POPE, $\Delta = -0.2$ on MME-RealWorld-Lite, $\Delta = -1.1$ on RealworldQA, $\Delta = -1.2$ on SEED, and $\Delta = -0.3$ on the test split of the VisCoT benchmark). This suggests Ground-R1 suffers from weak grounding consistency, often producing noisy evidence that fails to aid reasoning. Conversely, the observed gap validates our motivation: by shaping rewards via the factual-counterfactual uplift, CofactGVR incentivizes non-trivial causal gains, effectively mitigating shortcut reasoning.

Qualitative Results. In Figure 5, we provide representative qualitative examples to illustrate how CofactGVR performs evidence-grounded reasoning. We draw the following conclusions: (i) Evidence localization enables faithful visual confirmation. In the first example, CofactGVR identifies a compact region covering the racket and generates the answer conditioned on the cropped evidence. Notably, the intermediate trace explicitly states the intended evidence (*tennis*

racket) focuses on the corresponding object, providing an interpretable and verifiable rationale for the final decision. (2) Evidence is causally necessary for answering. In the second example, the queried company name is contained in a small region of the invoice, and CofactGVR localizes this region and extracts the answer directly from the cropped evidence. If the evidence is intervened—e.g., by replacing the predicted box with a random one and corrupting the resulting crop—the decisive text cue disappears. Please refer to Section A.7 for additional qualitative results, including analyses of failure cases. Furthermore, we discuss the current limitations and future outlooks in Section A.8.

5. Conclusion and Limitations

Conclusion. This paper presents **CofactGVR**, a reinforcement learning framework for GVR that explicitly quantifies the causal utility of generated evidence. We formalize GVR with a CGG that separates the shortcut reasoning path from the evidence-grounded path. Guided by CGG, CofactGVR adopts a two-stage pipeline: Evidence Generation stage produces candidate evidence along with a textual reasoning trace, and Counterfactual Answer Reasoning stage constructs counterfactual rollouts. Then, we propose counterfactual bonus shaping that rewards rollouts with a positive uplift over the counterfactual baseline. Finally, we introduce Q-PAS to filter low-signal rollouts and prioritize informative ones for improving sample efficiency. Extensive experiments across GVR benchmarks show consistent improvements, and evidence perturbation analyses reveal a larger gap between evidence preservation and intervention, confirming stronger evidence dependence. By bridging the counterfactual intervention with GVR, this framework provides effective learning signals for intermediate visual evidence, offering a fresh perspective to inspire broader advancements in faithful and interpretable MLLM reasoning.

Limitations. While CofactGVR significantly improves grounded visual reasoning, several limitations provide avenues for future work. First, although our method is theoretically agnostic to evidence modality, our current implementation relies empirically on bounding boxes; extending this to finer-grained formats like segmentation masks or keypoints remains an open challenge. Second, while the causal utility of visual evidence is explicitly quantified, the supervision of the accompanying textual reasoning traces remains implicit and could be enhanced via process reward models. Furthermore, our evidence acquisition primarily depends on cropping, which could be broadened into a tool-augmented, interactive process (e.g., multi-region selection, OCR) for complex real-world tasks. Finally, our framework currently lacks an adaptive mechanism to dynamically determine when grounding is necessary and how much reasoning computation to allocate based on query complexity.

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Impact Statement

This paper advances grounded visual reasoning by introducing counterfactual learning signals that encourage MLLM to rely on causally useful visual evidence. Our method may benefit applications that require transparent decision-making, such as document understanding and safety-critical visual inspection, where users need to verify what evidence supports a prediction. At the same time, the improved grounding and reasoning capabilities could be misused for generating convincing but misleading analyses, enabling privacy-invasive interpretation of sensitive imagery, or amplifying existing dataset biases if deployed without appropriate safeguards.

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A. Appendix

In this appendix, we further provide the following additional details, which are omitted in the main paper owing to the limited space. The structure of this Appendix is as follows:

- Section A.1 contains **additional methodological details**.
- Section A.2 contains **training configuration and implementation details**.
- Section A.3 contains **details of the evaluation benchmarks**.
- Section A.4 contains **analyses of training stability**.
- Section A.5 contains **additional ablation studies**.
- Section A.6 contains **extended related works**.
- Section A.7 contains **qualitative examples**.
- Section A.8 contains **limitations and outlooks**.

A.1. Additional Methodological Details

Algorithm Description for Q-PAS. The detailed procedure for Quantile-filtered Prioritized Advantage Sampling (Q-PAS) is presented in Algorithm 1. Q-PAS aims to mitigate the noise from samples with negligible advantages. First, we compute the importance score $s_i = |A_i|$ for each sample, acknowledging that both significantly good and bad outcomes provide valuable learning signals. Second, we apply a quantile filter (determined by hyperparameter q) to construct an effective set $\mathcal{S}$, pruning rollouts with trivial advantages. Finally, we perform prioritized sampling within $\mathcal{S}$ and obtain the selected subset $\mathcal{M}$, where the sampling probability p_i is weighted by the importance score.

Group-wise Advantage Construction. We follow the group-normalized advantage design in GRPO. For each training instance, we treat the $G = E \times A$ rollouts as one group and compute the normalized advantage:

$$A_{i,j} = \frac{\tilde{R}_{a,i,j} - \mu_{\tilde{R}}}{\sigma_{\tilde{R}}}, i \in [1, E], j \in [1, A], \quad (12)$$

where $\mu_{\tilde{R}}$ and $\sigma_{\tilde{R}}$ are the mean and standard deviation over the G shaped rewards within the same group.

GRPO Objective. Let π_θ be the current policy, $\pi_{\theta_{\text{old}}}$ be the rollout policy, and π_{ref} be a frozen reference model for KL regularization. For each selected rollout $(i, j) \in \mathcal{S}$, we define the importance ratio

$$\rho_{i,j}(\theta) = \frac{\pi_\theta(t_{i,j} \mid I, Q, e_i)}{\pi_{\theta_{\text{old}}}(t_{i,j} \mid I, Q, e_i)}. \quad (13)$$

We adopt the clipped GRPO surrogate with KL regularization:

$$\mathcal{L}_{\text{GRPO}}(\theta) = \mathbb{E}_{[Q, I, \{t_{i,j}\}_{i=1, j=1}^{E, A}] \sim \pi_{\theta_{\text{old}}}(t \mid Q, I)} \left[\frac{1}{M} \sum_{(i,j) \in \mathcal{M}} \min \left(\rho_{i,j}(\theta) A_{i,j}, \text{clip}(\rho_{i,j}(\theta), 1 - \epsilon, 1 + \epsilon) A_{i,j} \right) - \beta \text{KL}(\pi_\theta \parallel \pi_{\text{ref}}) \right], \quad (14)$$

where ϵ is the clipping range and β controls the strength of KL regularization. The policy parameters are updated by minimizing $\mathcal{L}_{\text{GRPO}}$.

Algorithm 1 Quantile-filtered Prioritized Advantage Sampling (Q-PAS) for GRPO

Require: Group-wise advantages $\mathbf{G} = \{A_{i,j}\}_{i=1;j=1}^{E,A}$ ($G = E \times A$), rollout tensors $\mathcal{T}$, quantile level q , prioritization exponent $\alpha > 0$, sampling ratio ρ

Ensure: Selected rollouts $\mathcal{T}_{\text{sel}}$ and advantages $\mathbf{A}_{\text{sel}}$

- 1: Compute importance scores: $s_i \leftarrow |A_i|$
- 2: Compute quantile cutoff: $\omega \leftarrow \text{Quantile}(\{s_i\}_{i=1}^G, q)$
- 3: Effective set: $\mathcal{S} = \{(i, j) \mid s_{i,j} \geq \omega\}$
- 4: Compute the prioritized distribution for $i \in \mathcal{S}$: $p_i \leftarrow \frac{(s_i)^\alpha}{\sum_{j \in \mathcal{E}} (s_j)^\alpha}$
- 5: Number of selected samples: $M \leftarrow \rho \times G$
- 6: Sample index set $\mathcal{M}$ from $\mathcal{S}$ according to p (with replacement)
- 7: Select subsets: $\mathcal{T}_{\text{sel}} \leftarrow \mathcal{T}[\mathcal{M}]$, $\mathbf{A}_{\text{sel}} \leftarrow \mathbf{A}[\mathcal{M}]$
- 8: Return $\mathcal{T}_{\text{sel}}$, $\mathbf{A}_{\text{sel}}$

Table 6. Implementation details and hyperparameters.

Configuration	Value	Configuration	Value
Backbone model	Qwen2.5-VL-7B-Instruct	Total rollouts (G)	8
Max prompt length	8192	Temperature	1
Max completion length	512	Random-box sampling $p_{\text{rand}}(\cdot \mid I)$	Uniform Distribution
Training steps	1000	Bonus trigger threshold (γ)	0
Global batch size	8	Bonus floor / cap ($b_{\text{min}}, b_{\text{max}}$)	0.2; 0.4
Learning rate	1e-6	Correctness-based mask threshold (τ)	0.99
Deepspeed config	ZeRO Stage 3	Top- k	3
Gradient accumulation	1	Quantile (q)	0.2
KL coefficient (β)	0.04	Prioritization exponent (α)	0.5
Reward type	Accuracy; Format	Sampling ratio (ρ)	1.5
Format reward weight (λ_{fmt})	2	Sampling with replacement	True
Evidence rollouts (E)	4	Compute resources	8 × A100-SXM4-80GB
Answer rollouts (A)	2	Training time	About 28 hours

A.2. Training Configuration

This subsection provides additional implementation details to facilitate reproducibility. We summarize the implementation details and training hyperparameters in Table 6, including the GRPO optimization configuration, counterfactual intervention parameters, and the proposed Q-PAS sampling hyperparameters. It should be noted that the counterfactual intervention branch is exclusively employed for reward estimation during the training phase. During inference, the model follows the standard factual path without constructing counterfactual rollouts, ensuring that CofactGVR incurs **no additional latency** compared to the standard two-stage pipeline. We further illustrate the exact prompt formats employed in our two-stage pipeline in Figure 6, covering the evidence-generation prompt and the answer-reasoning prompt.

A.3. Details of Evaluation Benchmarks

VisCoT (Shao et al., 2024a). We adopt VisCoT as a comprehensive grounded-reasoning benchmark that covers diverse vision–language skills under a unified evaluation protocol. Following prior work (Cao et al., 2025), we report results on the test split across five task families: Doc/Text understanding (DocVQA, TextCaps, TextVQA, DUDE, SROIE), Chart understanding (InfographicsVQA), General VQA (Flickr30k, Visual7W), Relation Reasoning (GQA, OpenImages, VSR), and Fine-grained VQA (CUB).

MME (Fu et al., 2025a). MME is a comprehensive benchmark for MLLMs, designed to assess the capabilities of model from multiple angles. It comprises 14 carefully curated subtasks that probe both low-level perception and higher-level reasoning. The benchmark is built around manually crafted instruction–answer pairs with intentionally concise prompts, which reduces spurious cues and promotes fairer comparisons across different models and training settings.

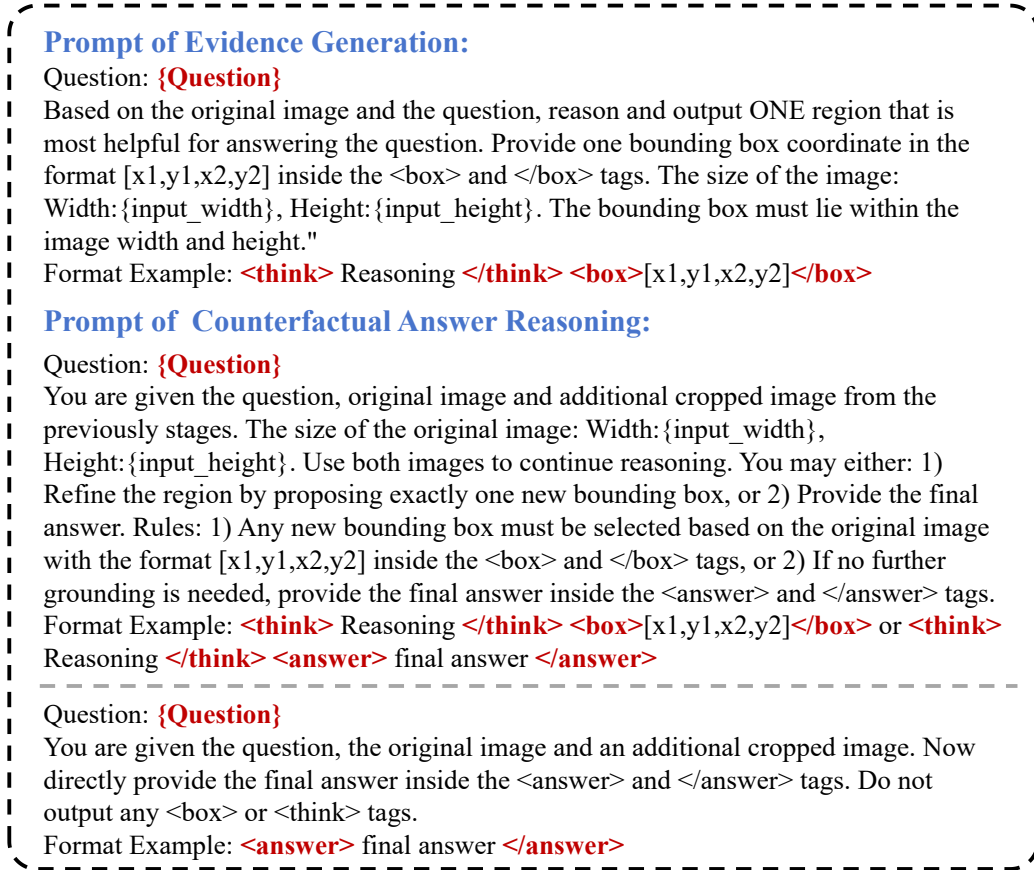


Figure 6. Prompts for CofactGVR.

POPE (Li et al., 2023d). POPE is a benchmark tailored to measure object hallucination in MLLMs, i.e., cases where a model asserts the presence of objects that are absent from the image. It contains 9,000 curated instances, each pairing an image with object-focused questions. The question set mixes positives (objects truly present) and deliberately constructed negatives (objects not present), enabling a controlled assessment of visual grounding and faithfulness under yes/no style queries.

SEED-Bench (Li et al., 2023a). SEED-Bench is a large-scale benchmark for evaluating the generative comprehension ability of MLLMs. It contains about 19K human-annotated multiple-choice questions, and covers 12 evaluation dimensions spanning both image and video understanding. SEED-Bench employs a dimension-targeted question generation pipeline with automatic filtering followed by manual verification, ensuring high-quality questions and options.

MME-RealWorld-Lite (Zhang et al., 2025b). MME-RealWorld is proposed to better reflect the practical difficulties that MLLMs encounter outside curated benchmarks. Different from the prior evaluations, MME-RealWorld collects a large set of high-quality, high-resolution images from public sources and the web, and pairs them with professionally annotated question-answer data spanning a range of real-world scenarios. The benchmark emphasizes challenging cases—often difficult even for humans—and aims to provide a more reliable, high-fidelity testbed for assessing real-world multimodal understanding and reasoning.

RealworldQA (xAI, 2024). RealWorldQA is a benchmark aimed at measuring multimodal models’ basic understanding of the physical world, with a particular focus on real-world spatial reasoning. It contains 700+ images, each paired with a question and a straightforward, verifiable answer. The images are drawn from authentic everyday settings, including anonymized vehicle-captured scenes as well as other real-world photos.

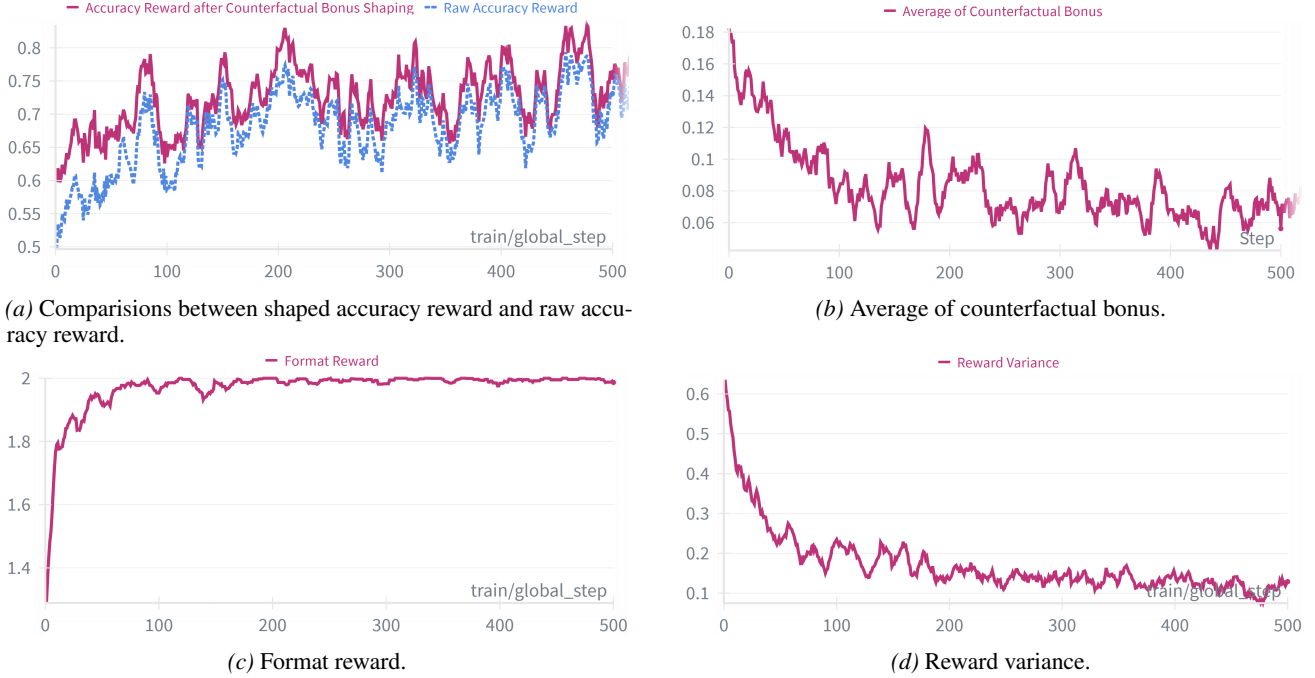


Figure 7. Visualization of training dynamics and reward statistics.

Visual Grounding Benchmarks. We evaluate the grounding performance on RefCOCO (Kazemzadeh et al., 2014), RefCOCO+ (Kazemzadeh et al., 2014), and RefCOCOg (Mao et al., 2016), which require localizing a referred region given a natural-language expression. Following common practice, we report accuracy under IoU-based correctness ($\text{IoU} \geq 0.5$) on the standard splits (RefCOCO: val/testA/testB; RefCOCO+: val/testA/testB; RefCOCOg: val/test). For MLLM-based grounding, we convert predicted boxes from model outputs to axis-aligned coordinates and compute IoU with ground-truth boxes. Following Ground-R1 (Cao et al., 2025), we leverage the following prompts as follows: *Locate <ref> in this image and output the bbox coordinates in JSON format.*

A.4. Training Stability Analysis

To better understand how CofactGVR learns under counterfactual bonus shaping, we visualize several training-time signals in Figure 7, including the accuracy reward before/after shaping, the magnitude of counterfactual bonus, the format reward, and the reward variance.

Shaped Accuracy Reward Consistently Improves the Optimization Signal. As shown in Figure 7a, the accuracy reward after counterfactual bonus shaping remains consistently above the raw accuracy reward throughout training. This indicates that the proposed counterfactual bonus provides a persistent and effective shaping signal, strengthening the learning gradients on rollouts whose evidence yields non-trivial causal gains. Meanwhile, the gap between the two curves stays bounded, suggesting that the trigger-and-clip design prevents reward inflation while still providing meaningful improvement.

Counterfactual Bonus Decays and Stabilizes as the Policy Becomes More Grounded. Figure 7b plots the average counterfactual bonus over training steps. The bonus starts relatively large and gradually decreases before stabilizing at a moderate level. This trend is expected: counterfactual interventions substantially degrade performance at the beginning of the training process, the large factual-counterfactual uplift leads to stronger bonuses. As the policy learns to produce more reliable evidence and grounded reasoning, the uplift becomes smaller and more stable, resulting in a lower but persistent bonus. Importantly, the non-zero plateau implies that counterfactual comparison continues to provide informative supervision even in later stages.

Format Reward Quickly Saturates. In Figure 7c, the format reward increases rapidly and saturates early, remaining stable thereafter. This suggests that format constraints are learned quickly and do not dominate the later optimization process.

Prompt of Single Prompt:

Question: {Question}

Based on the original image and the question, reason whether there exists a region in the image that could help you answer the question better. If such a region exists, provide one bounding box coordinate in the format [x1,y1,x2,y2] inside the <box> and </box> tags. The size of the image: Width:{input_width}, Height:{input_height}. The bounding box you provided should not exceed the image width and height. Then, you will receive a cropped image based on the bounding box. Use both images to continue reasoning inside a new <think> tag. You may conduct multiple rounds of grounding to refine your region as you want. The bounding box you provide should always be selected based on the original image. If at any point you determine no further visual information is needed, you may directly provide the final answer inside the <answer> and </answer> tags.

Format Example: <think> Reasoning </think> <box>[x1,y1,x2,y2]</box> OR
 <think> Reasoning </think> <answer> final answer </answer>

Figure 8. Single prompt with two stage instruction, which is employed by Ground-R1 (Cao et al., 2025).

Consequently, subsequent performance improvements are more likely attributed to evidence-aware reasoning behaviors.

Reward Variance Decreases, Indicating Improved Training Stability. Figure 7d shows that the reward variance drops sharply in the early stage and keeps decreasing over time. This reduced variance indicates a more stable reward landscape and more consistent policy updates, which aligns with our design choices that focus optimization on informative rollouts while suppressing noisy updates.

Overall, these curves corroborate the core intuition of CofactGVR: counterfactual bonus shaping provides an effective yet controlled learning signal for evidence-based reasoning, and the optimization remains stable and well-behaved.

A.5. Additional Ablation Studies

Impact of the Prompt Strategy on Training Stability and Effectiveness. To validate the necessity of our two-stage prompting design (i.e., separate prompts for Evidence Generation and Answer Reasoning, as shown in Figure 6), we compare it against ‘Single Prompt’ baseline where the model is implicitly instructed to generate reasoning traces and final answer. The prompt is shown in Figure 8. As illustrated in Figure 9, the Single Prompt strategy (green curves) exhibits significant instability. Specifically, Figure 9b reveals that the Single Prompt approach struggles to consistently trigger the counterfactual bonus mechanism, leading to frequent collapses in the bonus signal (Figure 9a and Figure 9c). In contrast, our Stage-specific Prompt strategy (pink curves) effectively decouples the generation tasks, maintaining a high trigger probability and providing a stable, dense learning signal that translates into superior accuracy rewards (Figure 9d).

Analysis of Bonus Evolution and Convergence. As observed in Figure 9 (pink curves), the mean bonus, total bonus, and trigger probability exhibit a characteristic trend: an initial decline followed by stabilization. This dynamic reflects the two-phase optimization process of CofactGVR:

- **Phase I: Pruning Spurious Evidence.** In the early stages of training, the policy explores the action space by generating a large number of candidate regions, resulting in high initial trigger probabilities and total bonus values. However, many of these proposed regions are spurious—they are visually salient but do not causally contribute to the final answer (i.e., intervening on them does not change the prediction). As the counterfactual mechanism penalizes these causally ineffective regions (via low or zero utility scores), the model learns to “prune” them. This leads to a temporary decrease in the quantity of generated evidence, causing the drop in total and mean bonus metrics.
- **Phase II: Causal Alignment and Stabilization.** As training progresses, the policy converges towards faithful reasoning. It learns to generate visual evidence only when it is strictly necessary for deriving the correct answer. The stabilization of the metrics indicates that the model has reached an equilibrium where it reliably produces high-quality, causally significant evidence. The stabilization of the trigger probability (Figure 9b) further confirms that the model has internalized the grounding format as a functional tool for reasoning, rather than merely satisfying a syntactic constraint.

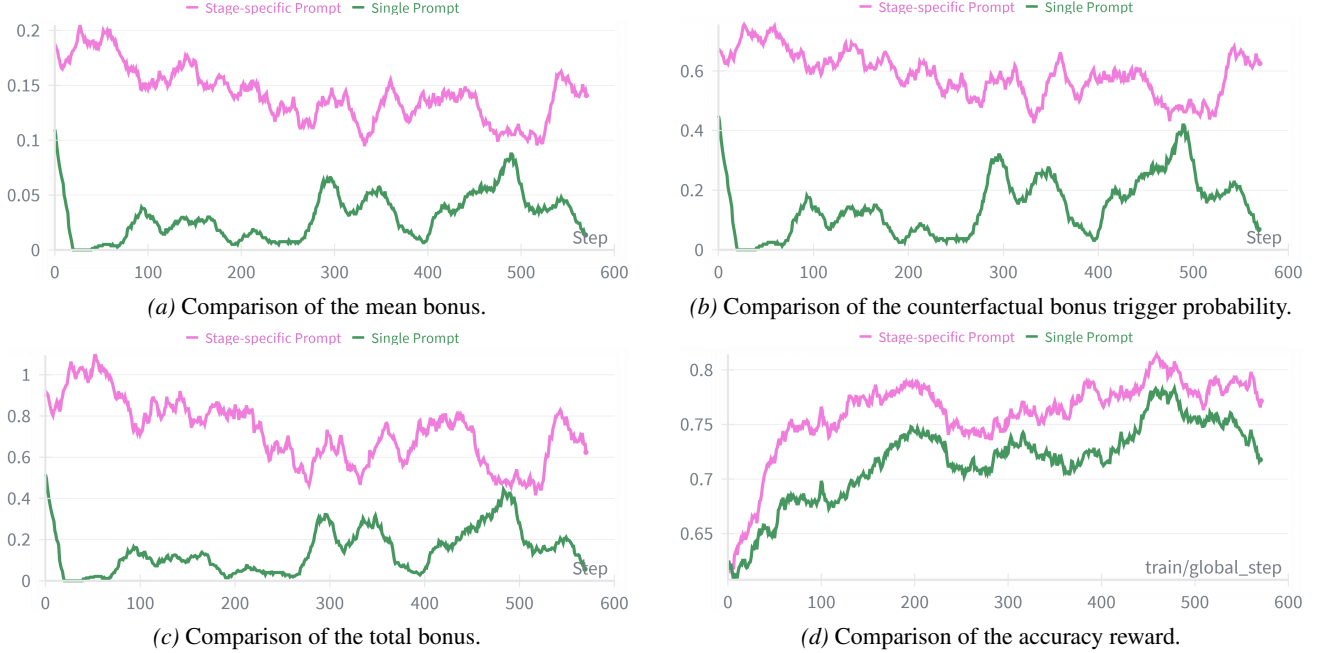


Figure 9. Training dynamics comparison between Stage-specific Prompts (Ours) and a Single Prompt baseline.

Why Shaping only Accuracy Reward? We deliberately compute the uplift and bonus shaping *only* on accuracy rewards for two reasons. On the one hand, our goal is to quantify the *causal utility of visual evidence*; accuracy is the task-relevant outcome that directly reflects whether the predicted evidence improves grounded reasoning, whereas format reward is largely evidence-agnostic (it mainly checks output structure) and would therefore confound the causal attribution if included in the factual-counterfactual gap. On the other hand, applying shaping to the counterfactual branch is conceptually unnecessary: the counterfactual rollouts serve as a *controlled baseline* to estimate uplift, and altering their rewards would change the baseline itself, weakening the interpretability and stability of the reward gap.

A.6. Extended Related Works

Visual Reasoning in MLLMs Recent progress in MLLMs (Shao et al., 2024a; Wang et al., 2025; Xu et al., 2025a) has sparked sustained interest in endowing models with stronger visual reasoning abilities, i.e., solving problems through explicit multi-step trajectories that are both accurate and interpretable. Early efforts largely rely on prompting to elicit rationales, using carefully designed reasoning templates (Yao et al., 2023) or structured search strategies (Besta et al., 2024) to mitigate the limitation of single-path chain-of-thought (CoT). Beyond prompting, a broad line of work improves reasoning via supervised fine-tuning (SFT) on CoT-style annotations (Zhang et al., 2024d; Wang et al., 2024a; Dong et al., 2025; Xu et al., 2025b; Yao et al., 2025; Thawakar et al., 2025), where reasoning stages are explicitly defined to enhance process interpretability. However, such CoT-driven SFT pipelines typically depend on large-scale rationale annotations and may suffer from limited generalization or amplified language bias when the reasoning text drifts away from visual evidence (Trung et al., 2024; Cao et al., 2025).

Motivated by the success of reasoning-centric LLMs (Shao et al., 2024b; Guo et al., 2025), recent studies have turned to RL for post-training, showing that RL could stimulate behaviors such as self-correction and iterative refinement without relying solely on curated CoT annotations. This RL-inspired paradigm has rapidly expanded to multimodal settings, leading to a series of RL-based reasoning MLLMs (Yang et al., 2025; Peng et al., 2025; Tan et al., 2025; Liu et al., 2025; Meng et al., 2025; Yu et al., 2025; Huang et al., 2026). Nevertheless, many existing approaches still treat the image primarily as an input condition, and conduct most of the reasoning solely in the textual space, which makes it difficult to guarantee that intermediate thoughts are grounded in verifiable visual evidence. A complementary direction explicitly integrates visual evidence into the reasoning process. Representative methods (Cao et al., 2025; Fan et al., 2025; Zhang et al., 2025a; Wang et al., 2026; Sarch et al., 2025) collect or predict bounding boxes of key regions, interleave these localized observations with textual reasoning traces, and re-answer questions based on cropped regions in a multi-round generation procedure.

In contrast, our work focuses on causally grounded visual reasoning. We view GVR as an evidence-centric process in which intermediate variables (e.g., predicted evidence and reasoning traces) mediate the final prediction, and formalize this perspective via a Causal Grounding Graph (CGG). Building on the two-stage “grounding-then-answering” paradigm, we further introduce counterfactual rollouts that directly intervene on the predicted evidence and turn the factual-counterfactual uplift into a principled training signal. This design explicitly discourages shortcut reasoning that bypasses evidence and encourages the model to generate evidence that is not only plausible, but causally useful for answering.

A.7. Qualitative Examples

We showcase representative reasoning examples on multiple benchmarks, including predicted evidence regions, and model outputs. As shown in Figure 10–Figure 15, CofactGVR typically produces explicit visual evidence in the reasoning trace and answers the question conditioned on the corresponding cropped images, yielding an interpretable evidence-to-answer chain. For instance, on general VQA and real-world understanding tasks, the model localizes task-relevant objects or sign/text regions and leverages the cropped image to confirm the final decision (Figure 10, Figure 13, Figure 14). These examples suggest that the model actively seeks localized evidence rather than relying purely on the global image or shortcut reasoning.

Failure Cases and Discussions. We also present challenging examples in Figure 15. On InfographicsVQA, the model may still fail on text-heavy, densely structured images, producing an incorrect answer even though the key information exists in the image. On MME-RealWorld-Lite, we observe occasional trace-evidence misalignment: the final answer is correct, while the predicted region and the corresponding explanation do not faithfully support it. These cases indicate that improving task performance does not automatically guarantee perfectly faithful grounding, especially in visually cluttered scenes or text-centric settings.

A.8. Limitations and Outlooks

While our approach demonstrates strong improvements and improves evidence dependence via counterfactual intervention training, several limitations remain and motivate future research.

Alternative Evidence Formats. While CofactGVR currently relies on bounding boxes, our Causal Grounding Graph (CGG) is theoretically agnostic to the evidence format and could naturally extend to other parameterizable modalities via corresponding *do*-operator interventions (e.g., segmentation masks, keypoints, or textual rationales). We empirically adopt bounding boxes since they inherently provide both location and scale to handle diverse object scale variations and current MLLMs heavily depend on adaptive cropping to perceive high-resolution details. Extending our reinforcement learning pipeline to support these finer-grained alternative modalities remains a promising direction for future research.

Granularity of Reasoning Supervision. While CofactGVR successfully quantifies the causal utility of the visual evidence, the supervision of the accompanying *textual* reasoning trace remains implicit. Currently, the model is incentivized to generate causally necessary visual regions, but the semantic consistency of the generated text describing these regions is not explicitly enforced. Future work could explore integrating fine-grained process reward models (PRMs) to ensure that the textual explanation is as causally robust as the visual evidence.

Beyond Cropping: Tool-augmented and Interactive “Think-with-Image” Reasoning. CofactGVR currently relies on cropping as the primary mechanism for evidence acquisition. While effective, cropping alone is often insufficient for real-world problems that require reading dense text, comparing multiple distant regions, or using external knowledge. A natural next step is to equip the model with lightweight tools—such as multi-region selection, zooming, OCR, or retrieval—and allow it to alternate between *thinking* and *acting*: selecting a tool, gathering new evidence, and then updating its reasoning based on the returned results. Such a tool-augmented and interactive setting could better realize “think with image” reasoning beyond a single crop.

When Ground and How Much Reason. Not all questions require the same level of grounding or reasoning depth. Simple questions could be solved with minimal evidence and short reasoning, while others demand multi-step exploration and verification. Recent studies (Sarch et al., 2025; Wei et al., 2022) have likewise highlighted this issue. A key limitation of current pipelines is the lack of explicit mechanism to decide *when* to ground is necessary and *how much* to reason. Future work may explore adaptive policies that allocate computation based on difficulty and uncertainty, e.g., early exiting for easy queries, deeper evidence search for hard cases, and learned stopping criteria.

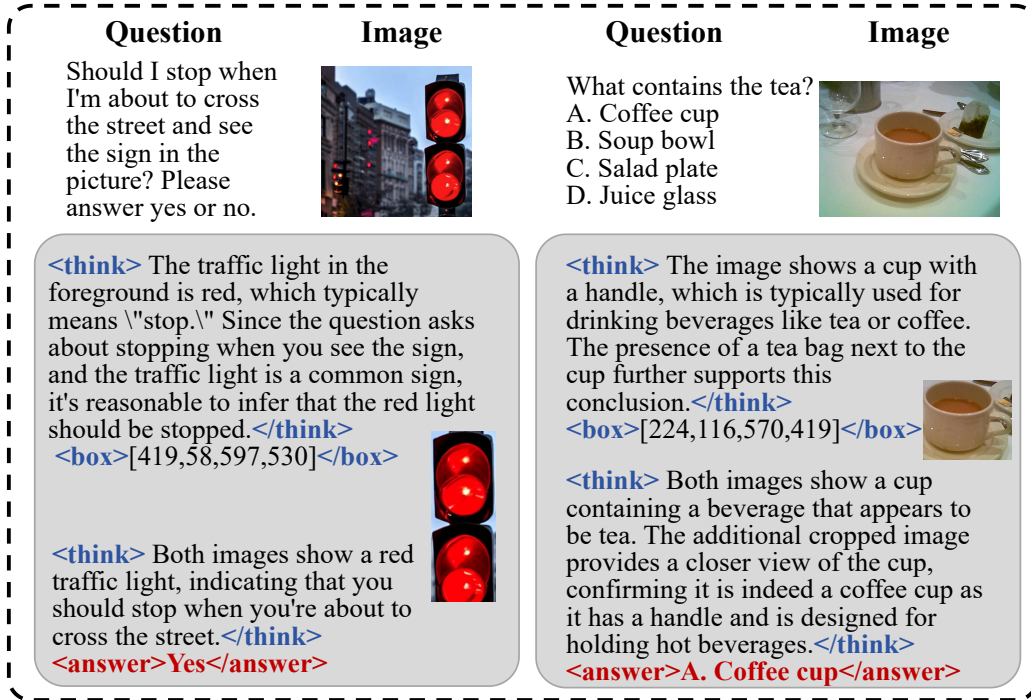


Figure 10. Reasoning trajectories of our CofactGVR on MME (left) and Openimage (right) benchmarks.

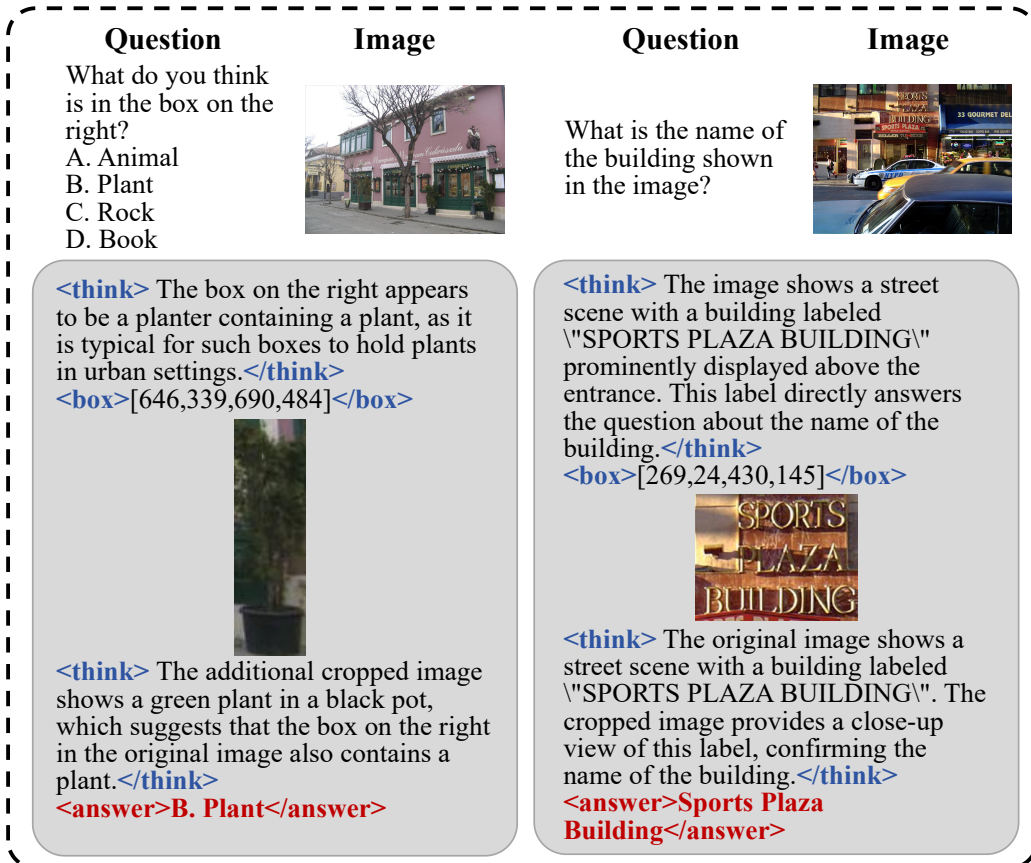


Figure 11. Reasoning trajectories of our CofactGVR on GQA (left) and TextVQA (right) benchmarks.

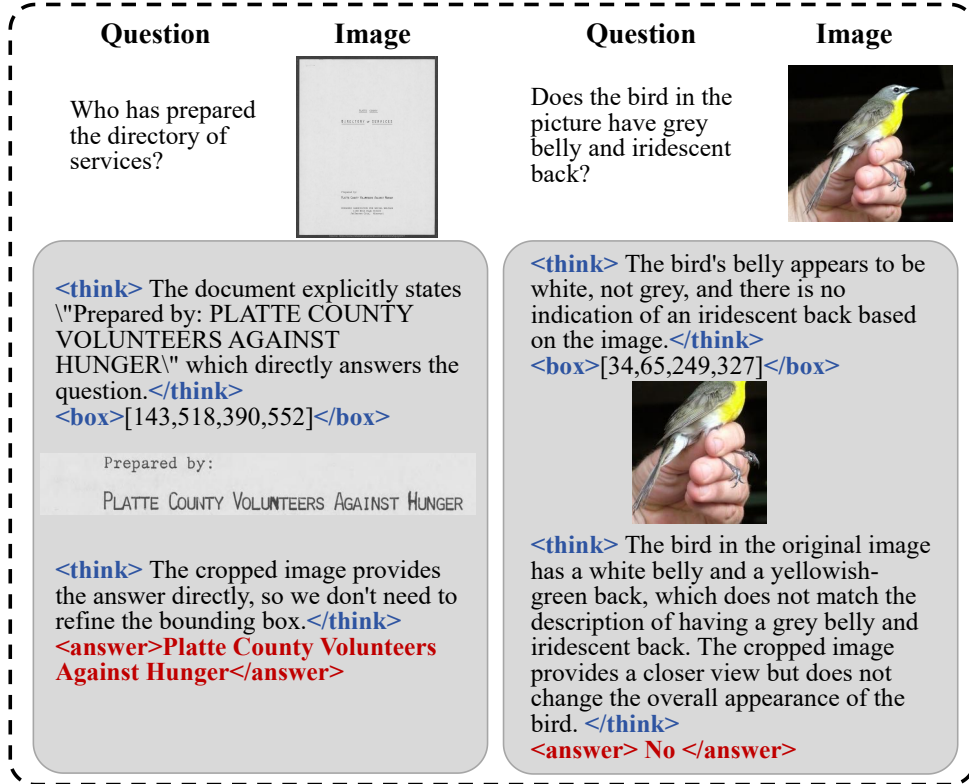


Figure 12. Reasoning trajectories of our CofactGVR on DocVQA (left) and CUB (right) benchmarks.

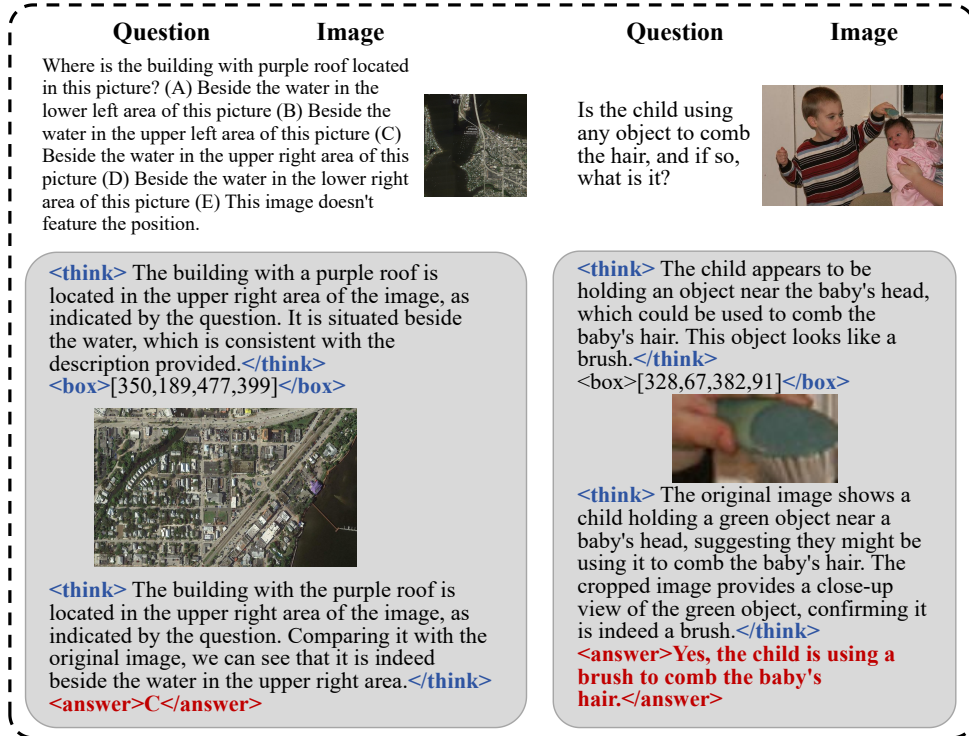


Figure 13. Reasoning trajectories of our CofactGVR on MME-RealWorld-Lite (left) and Flickr30K (right) benchmarks.

Question

This image shows the front view of the ego car. What is the status of the cars that are to the front of the ego car?

(A) Many cars are moving, and many are parked.
 (B) Two of the cars are moving, and many are parked.
 (C) Many cars are moving, and three are parked.
 (D) Three of the cars are moving, and many are parked.
 (E) The image does not feature the object.

Image



Reasoning Trajectory:

<think> The image shows a street with several cars parked along both sides. There are a few cars visible in motion, but the majority appear to be parked. The question asks about the status of the cars in front of the ego car, which is not specified in the image. However, based on the options provided, we can infer that there are multiple cars moving and parked in front of the ego car. **</think>**

<box>[307,239,379,305]**</box>**



<think> The original image shows a street scene with multiple cars parked along both sides. The additional cropped image appears to focus on a different part of the scene, possibly highlighting a specific area or object. However, it does not provide any new information about the status of the cars in front of the ego car. Since the original image clearly shows a street with parked cars, we can conclude that the cars in front of the ego car are likely parked as well. The cropped image does not change this conclusion. **</think>**

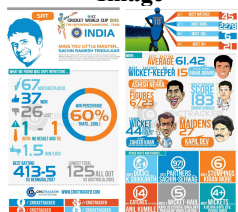
<answer>B**</answer>**

Figure 14. Reasoning trajectories of our CofactGVR on RealworldQA benchmark.

Question

Who has a record of the highest individual score by any Indian in the ICC cricket world cups so far?

Image



Reasoning Trajectory (Left):

<think> The image provides information about the highest individual score achieved by an Indian player in the ICC Cricket World Cup. It explicitly states that Sourav Ganguly holds the record with a score of 183. **</think>**

<box>[407,294,500,358]**</box>**



<think> The image highlights Sachin Tendulkar as the "Defending Champions" team India's player, and it mentions his record of the highest individual score by any Indian in the ICC cricket world cups so far. The cropped image does not add new information but reinforces the focus on Sachin Tendulkar. **</think>**

<answer>Sourav Ganguly**</answer>**

Question

How many cars are white in this picture? Please answer directly with a single word or number.

Image



Reasoning Trajectory (Right):

<think> There are several cars visible in the image, but only one car appears to be white. It is located near the center of the image, parked on the right side of the street. **</think>**

<box>[336,249,362,270]**</box>**



<think> The original image shows a street scene with multiple cars, including a white car. The cropped image focuses on a different part of the street, but it does not change the count of white cars present in the original image. **</think>**

<answer>2**</answer>**

Figure 15. Failed reasoning trajectories of our CofactGVR on InfographicsVQA (left) and MME-RealWorld-Lite (right) benchmarks. These two samples produce incorrect answers and reasoning traces, respectively.